Data Structure (Tree)

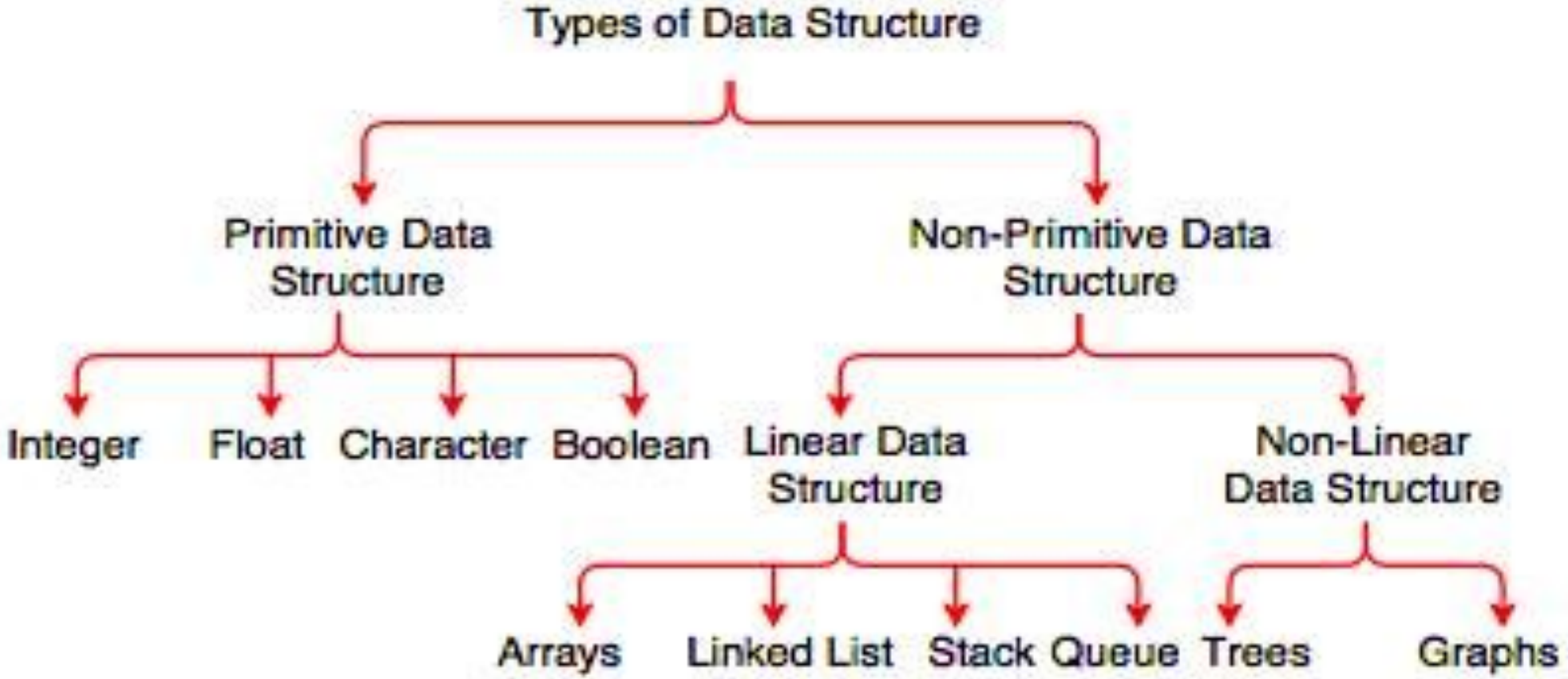
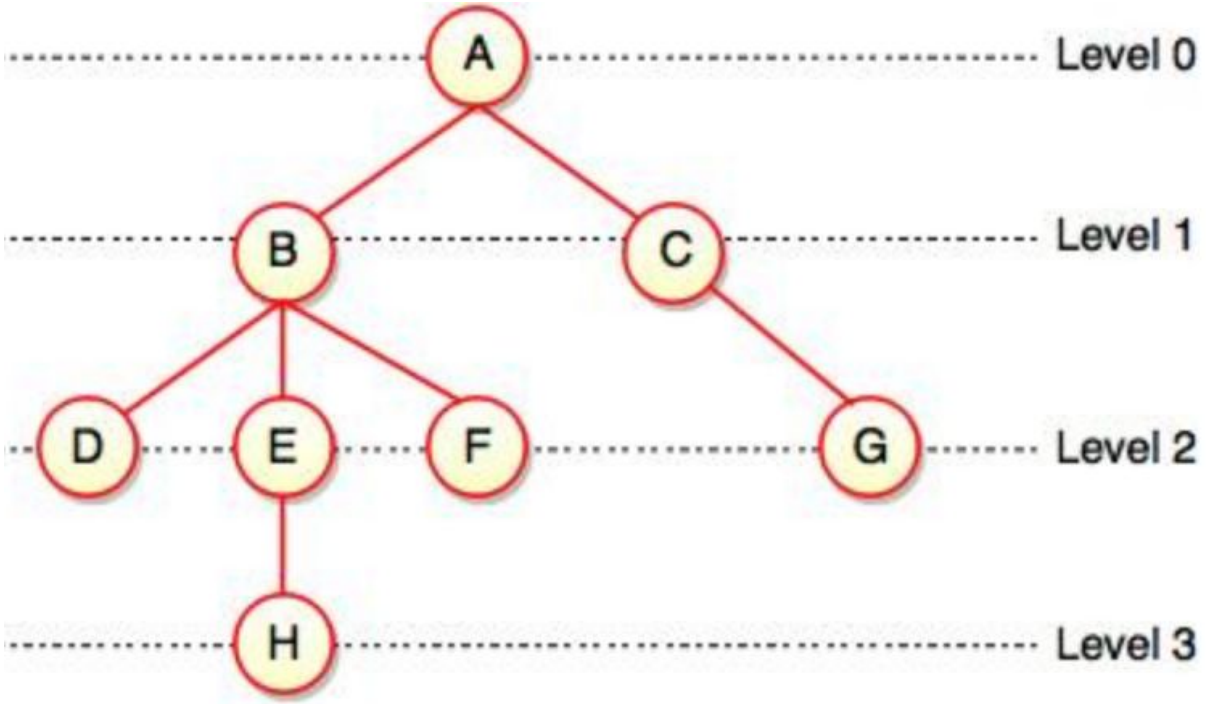
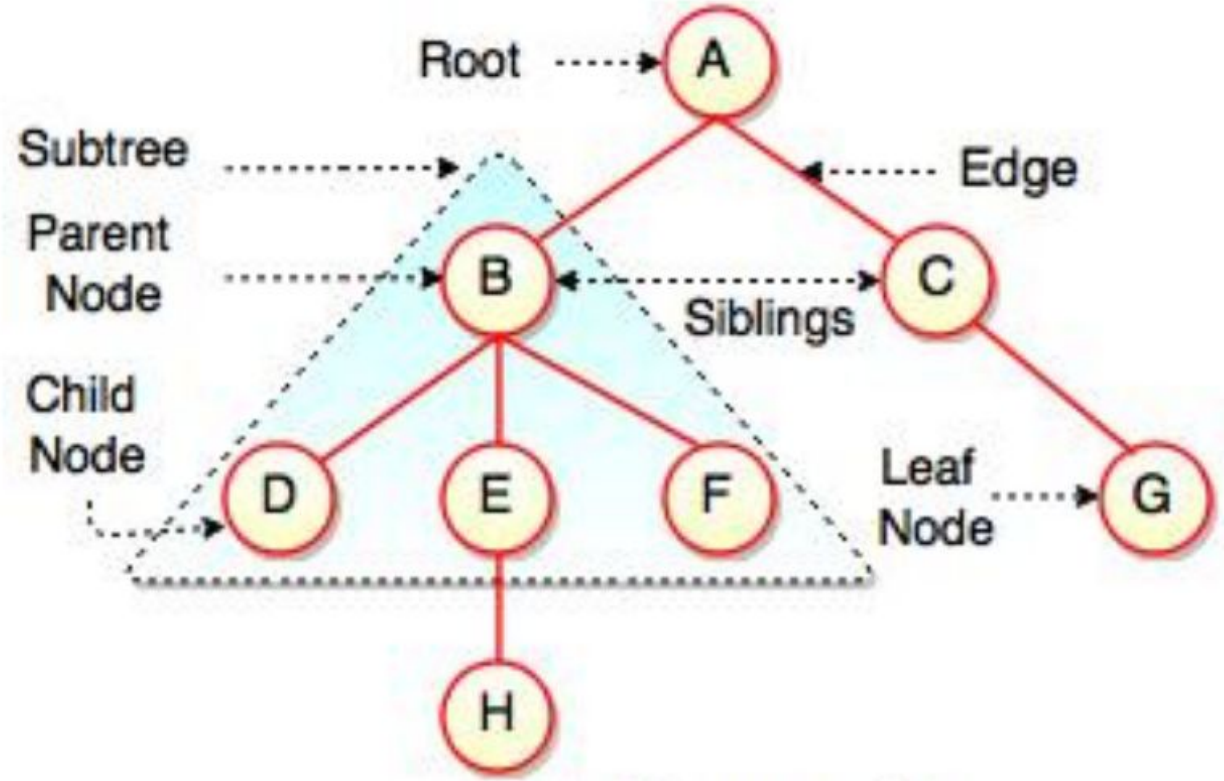


Fig. Types of Data Structure

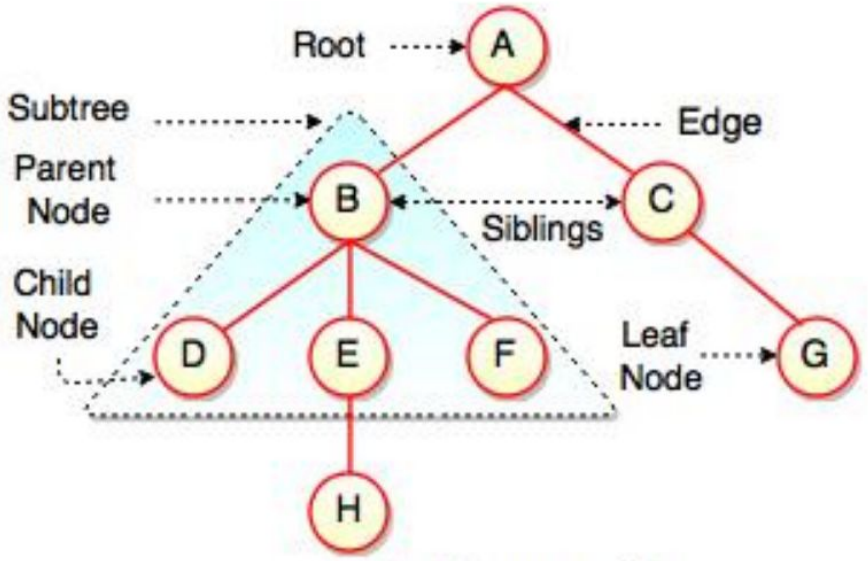
Tree Data Structure (Non Linear)

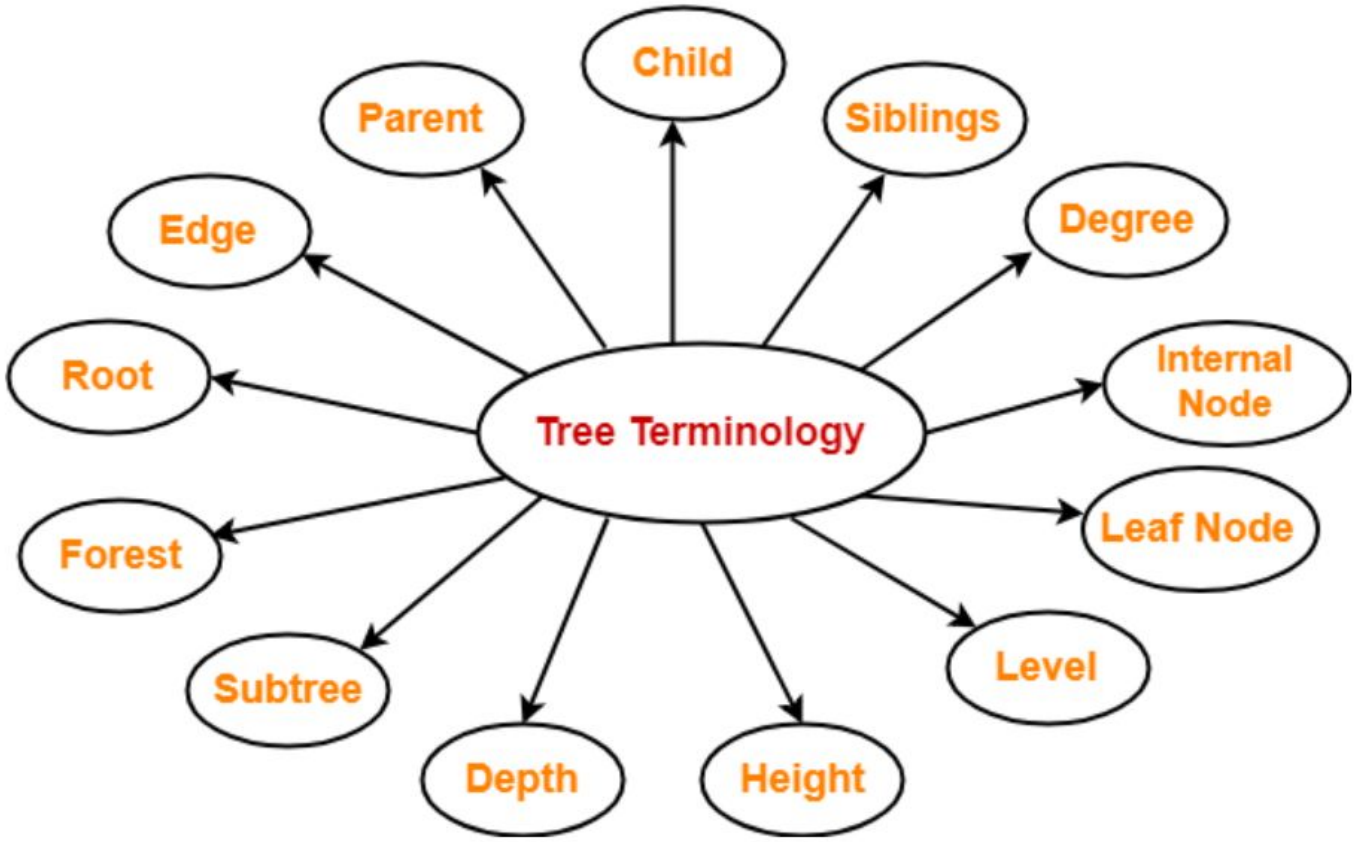
Tree is a hierarchical data structure which stores the information naturally in the form of hierarchy style. Tree is one of the most powerful and advanced data structures. It is a non-linear data structure compared to arrays, linked lists, stack and queue. It represents the nodes connected by edges.





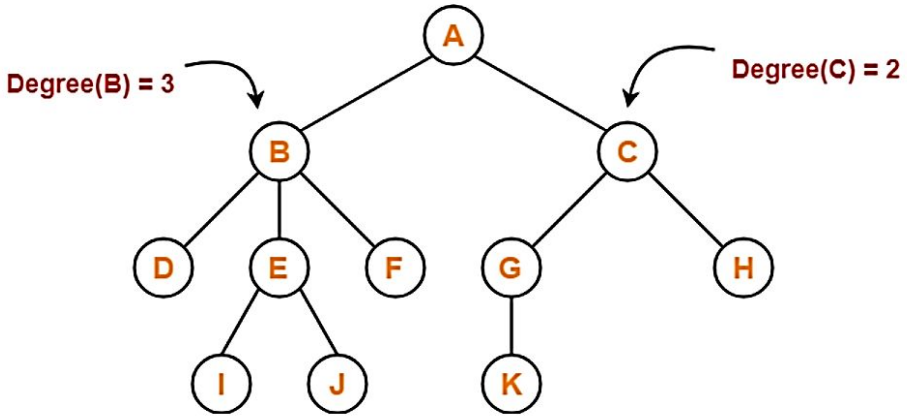
Tree Data Structure (Non Linear)



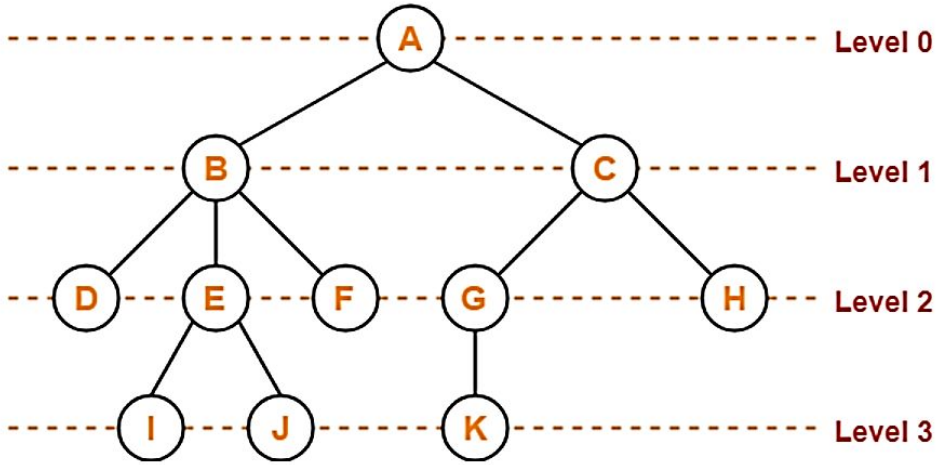


Tree Data Structure (terminologies)

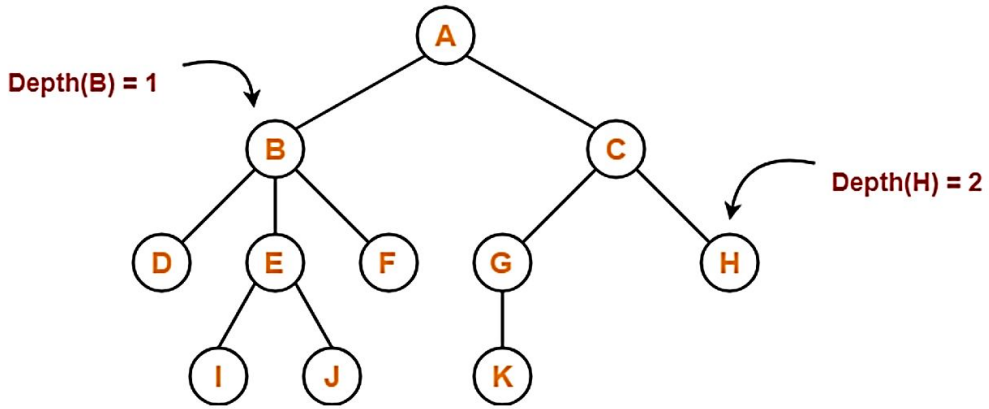
- **Degree of a node** is the total number of children of that node.
- **Degree of a tree** is the highest degree of a node among all the nodes in the tree.



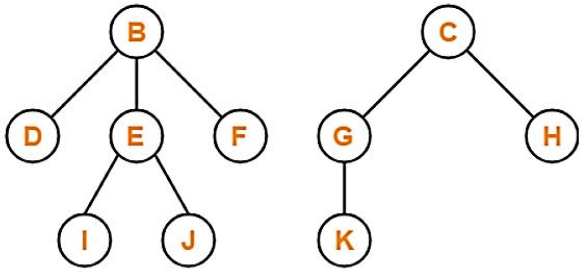
- In a tree, each step from top to bottom is called as **level of a tree**.
- The level count starts with 0 and increments by 1 at each level or step



- Total number of edges from root node to a particular node is called as **depth of that node**.
- **Depth of a tree** is the total number of edges from root node to a leaf node in the longest path, Depth of the root node = 0, The ter

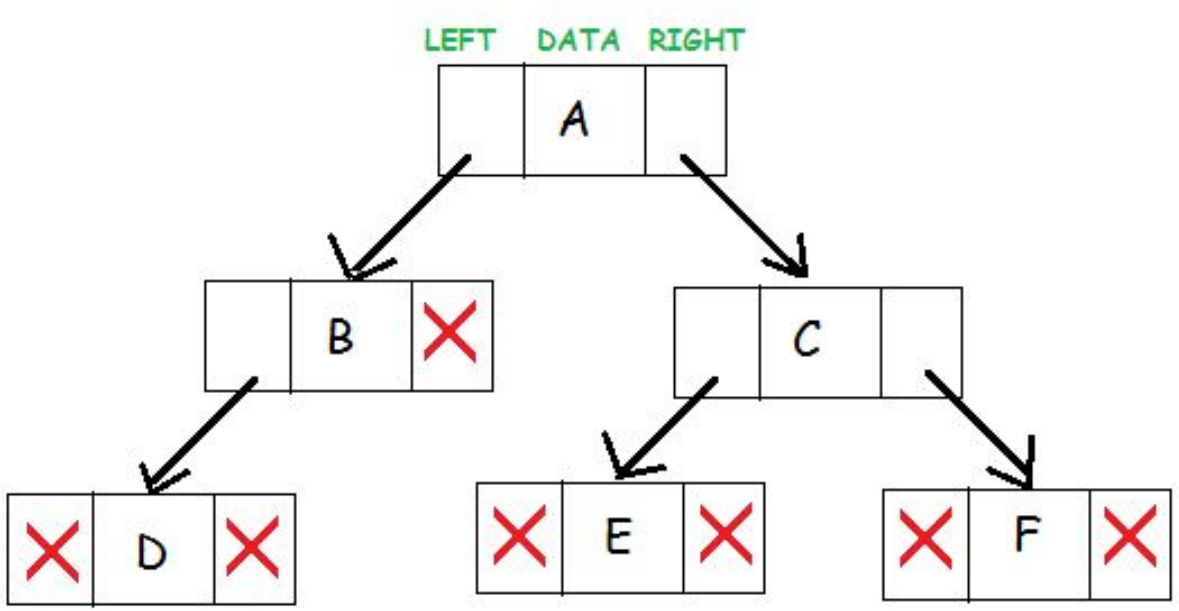


A forest is a set of disjoint tre



Forest

Tree Data Structure (Non Linear)



Tree in C language:

```
struct node
{
  int data;
  struct node *left;
  struct node *right;
};
```

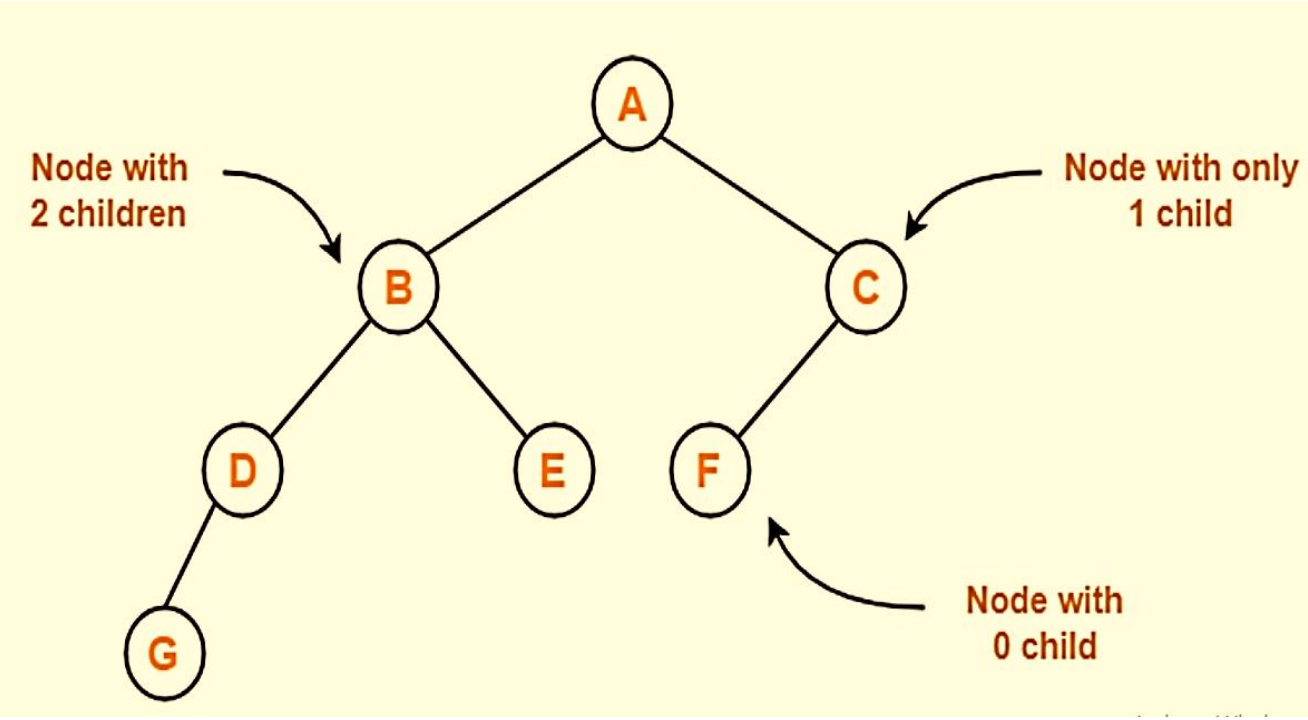
Tree Data Structure (Non Linear) Advantages

- Tree reflects structural relationships in the data.
- It is used to represent hierarchies.
- It provides an efficient insertion and searching operations.
- Trees are flexible. It allows to move subtrees around wit minimum effort.

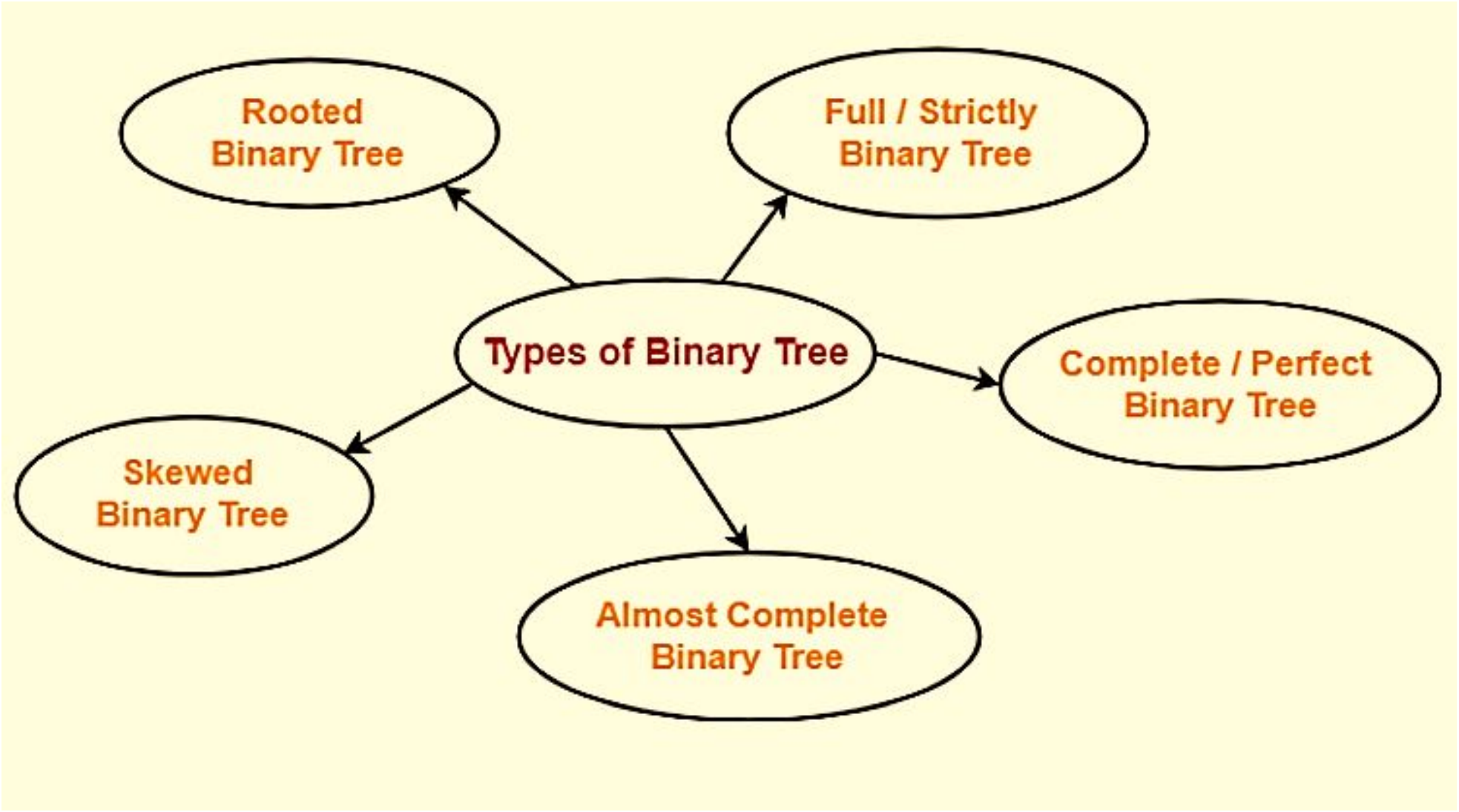
Tree Data Structure (Binary Tree)

Binary tree is a special tree data structure in which each node can have at most 2 children.

Thus, in a binary tree, Each node has either **0 child or 1 child or 2 children**.



Types of Binary Tree

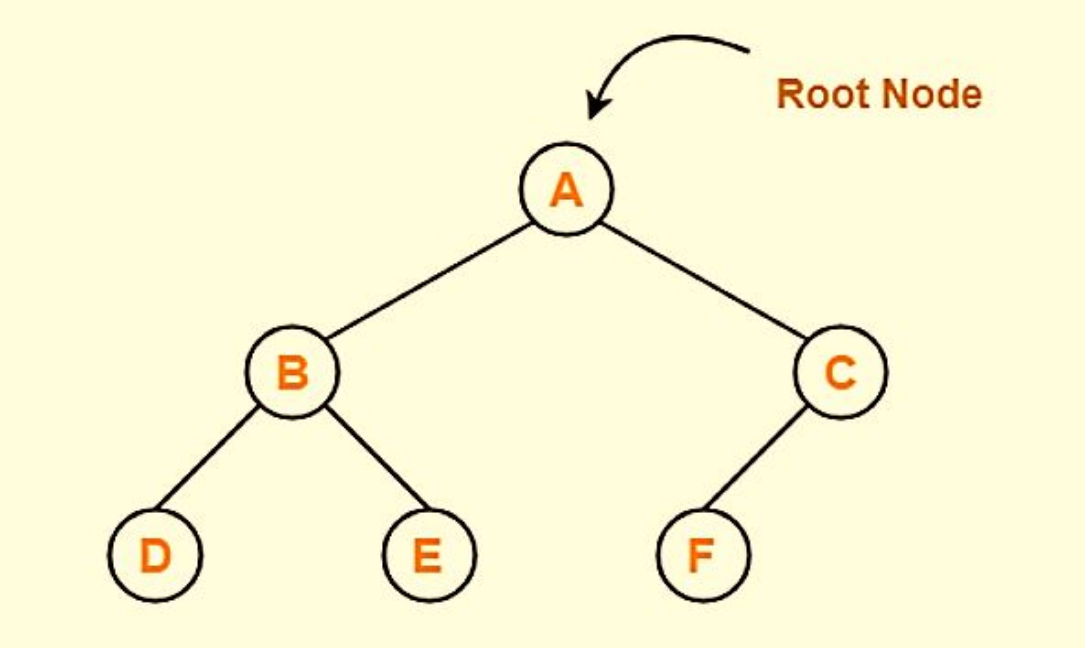


Types of Binary Tree

A **rooted binary tree** is a binary tree that satisfies the following 2 properties-

It has a root node.

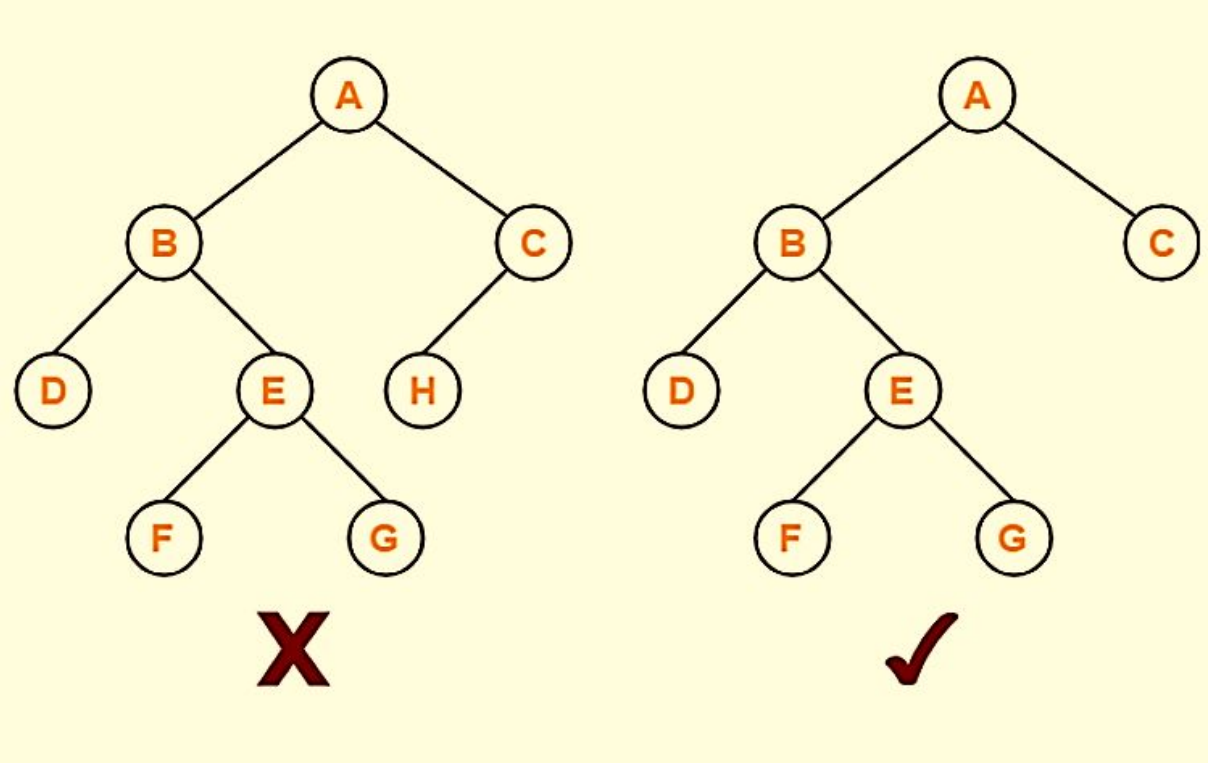
Each node has at most 2 children.



Types of Binary Tree

A binary tree in which every node has either 0 or 2 children is called as a **Full binary tree**.

Full binary tree is also called as **Strictly binary tree**.



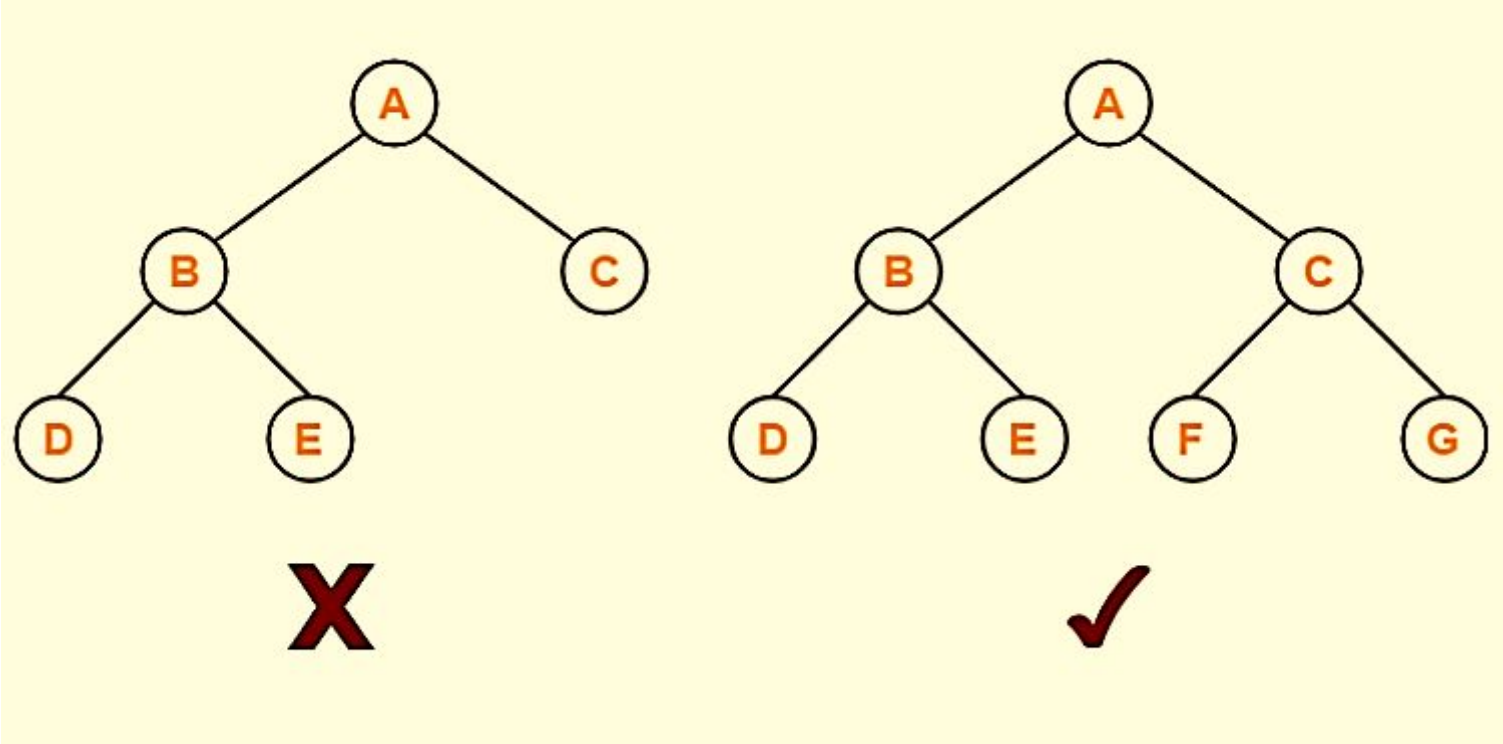
Types of Binary Tree

A **complete binary tree** is a binary tree that satisfies the following 2 properties-

Every internal node has exactly 2 children.

All the leaf nodes are at the same level.

Complete binary tree is also called as **Perfect binary tree**.

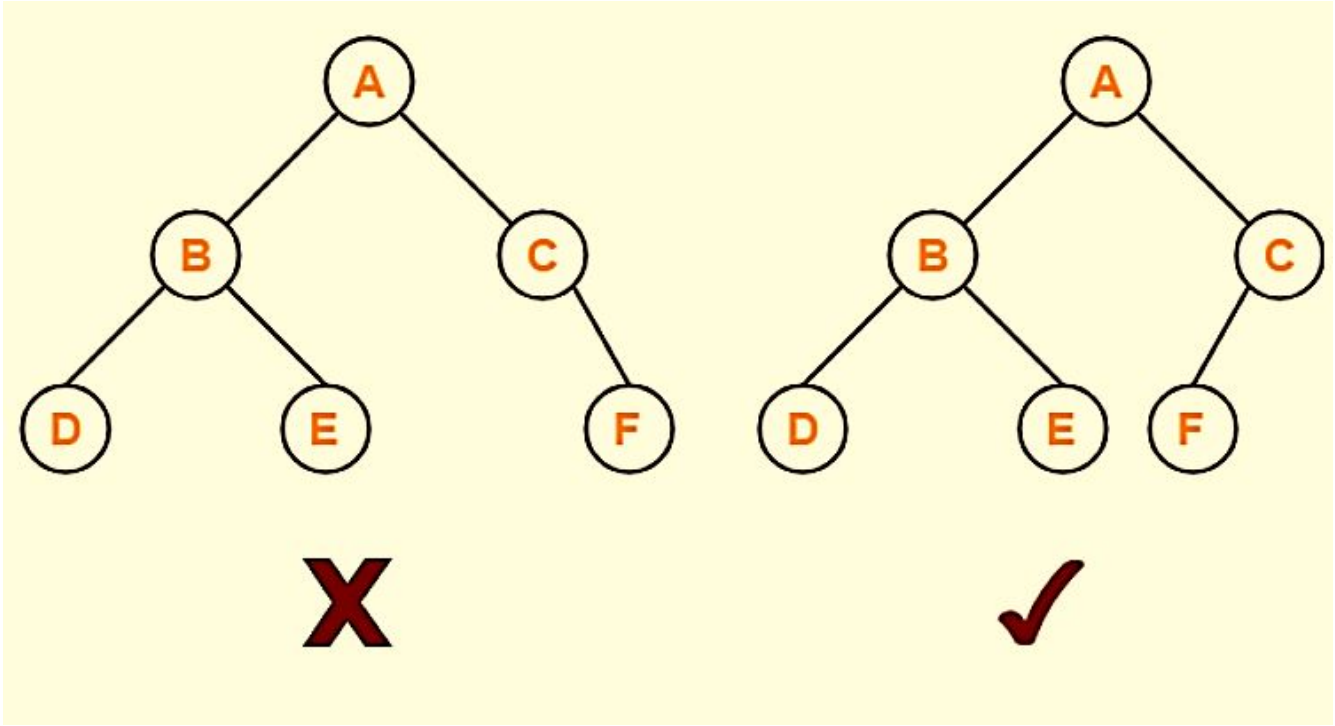


Types of Binary Tree

An **almost complete binary tree** is a binary tree that satisfies the following 2 properties:

All the levels are completely filled except possibly the last level.

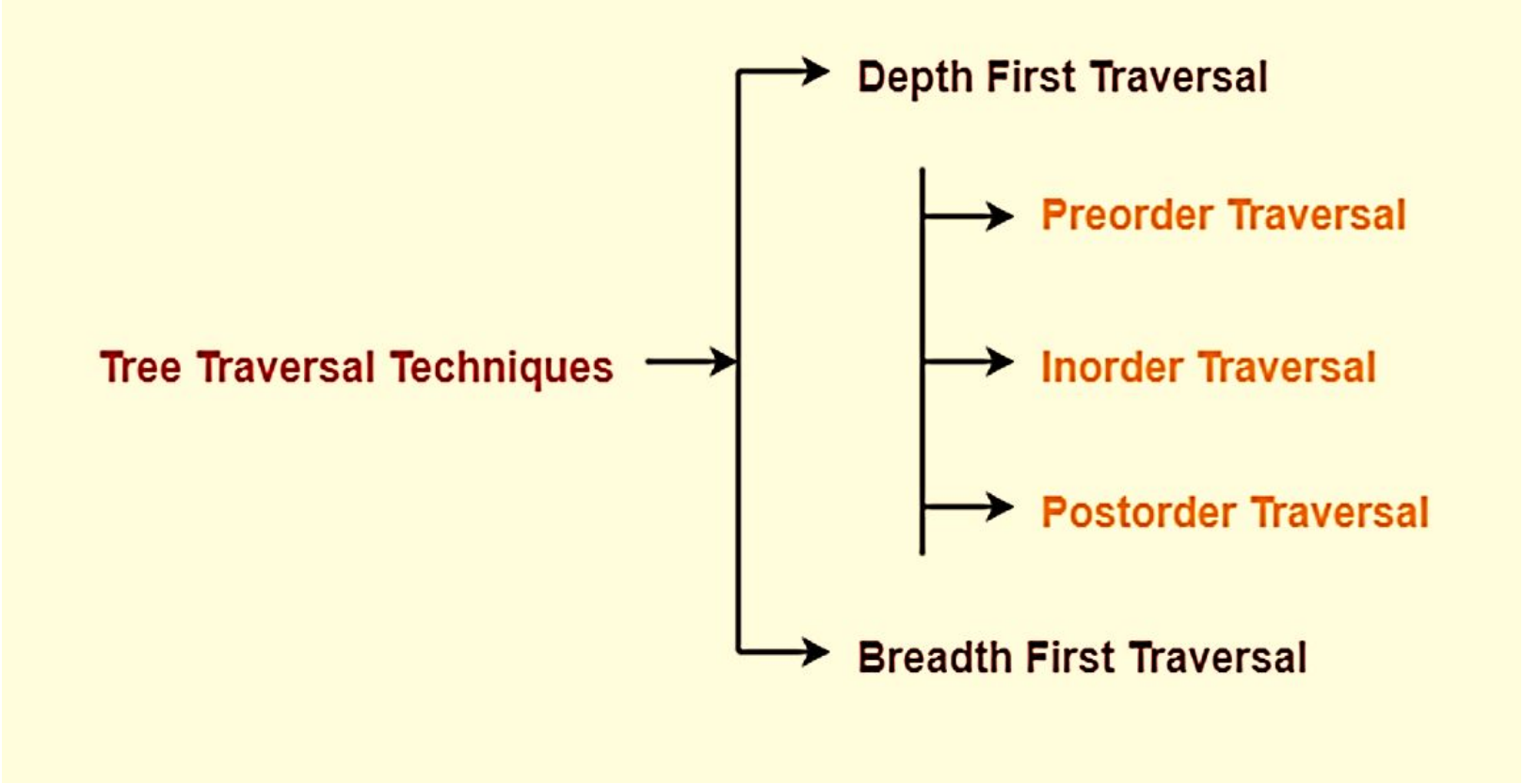
The last level must be strictly filled from left to right.



Binary Tree Traversal

Tree Traversal refers to the process of visiting each node in a tree data structure exactly once.

- Preorder Traversal
- Inorder Traversal
- Postorder Traversal



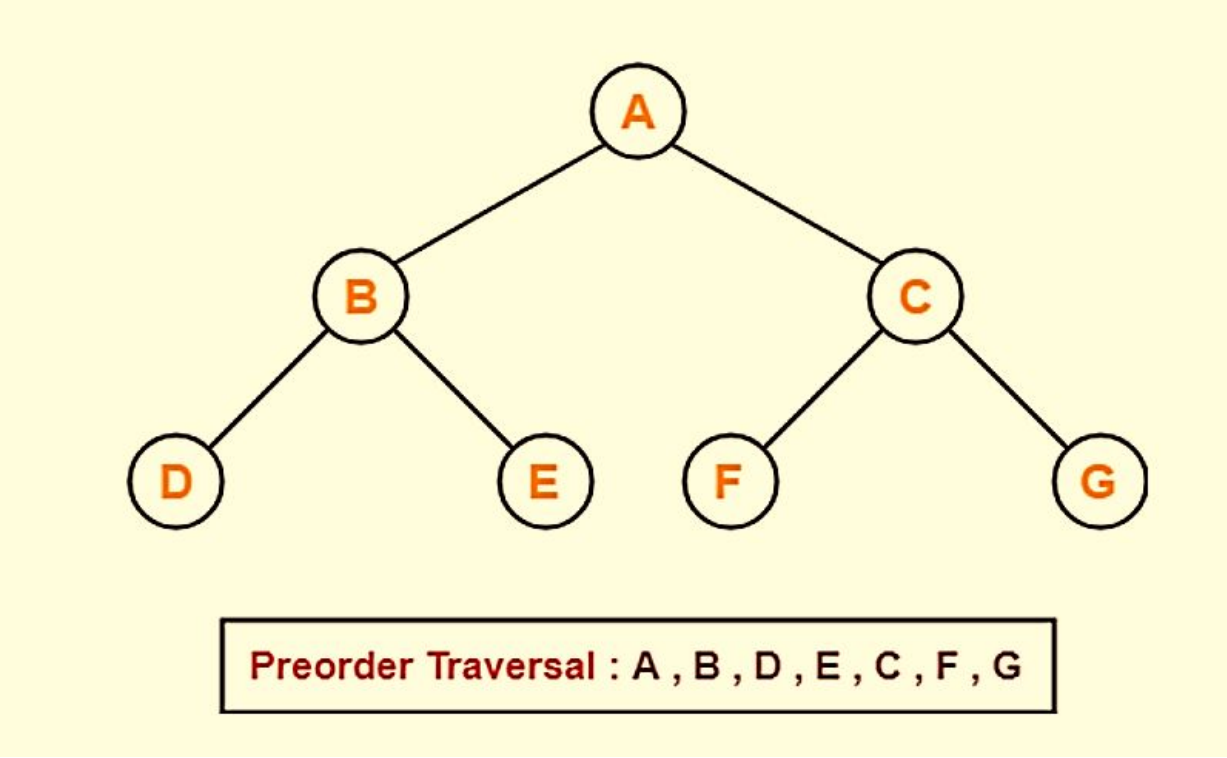
*Binary Tree Traversal (*Preorder Traversal)

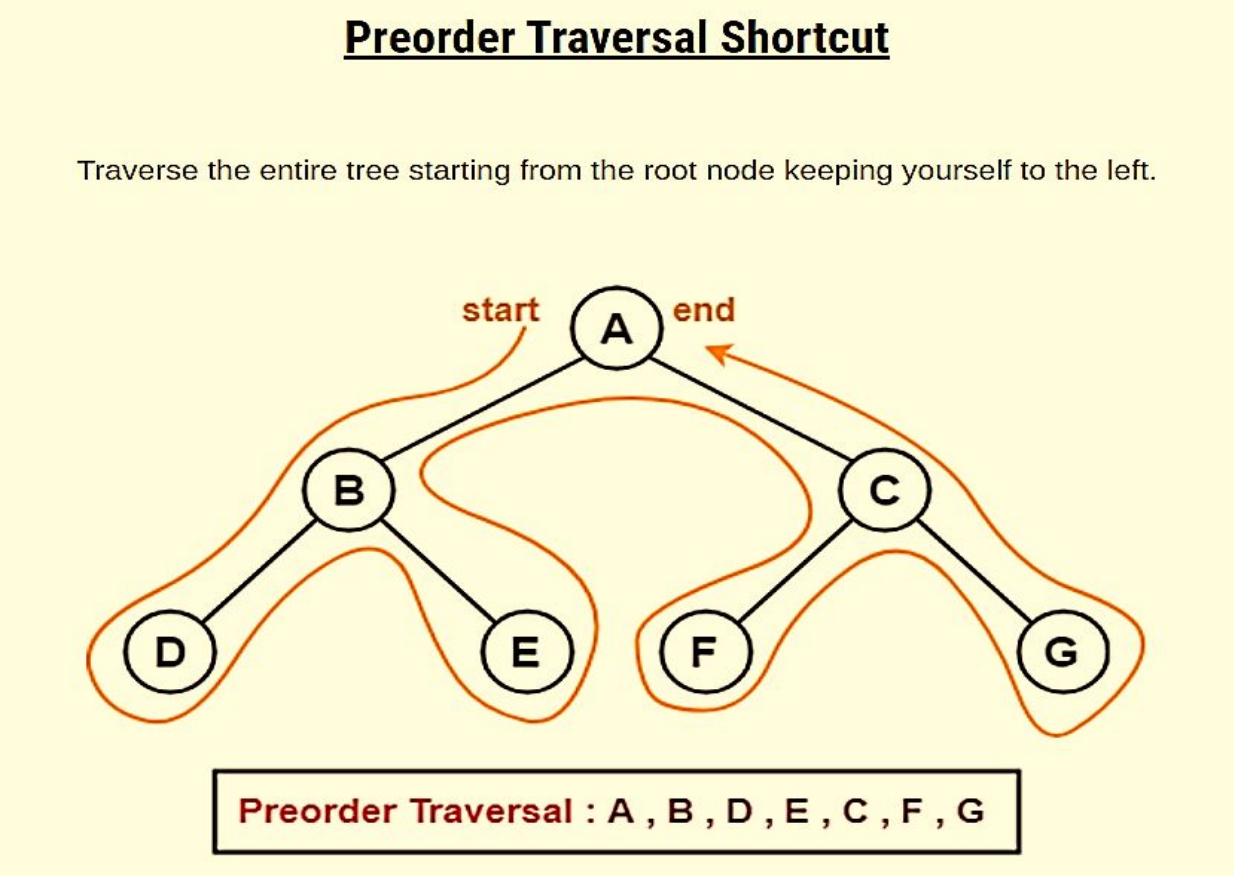
Visit the root

Traverse the left sub tree i.e. call Preorder (left sub tree)

Traverse the right sub tree i.e. call Preorder (right sub tree)

Root → Left → Right





- **Preorder traversal is used to get prefix expression of an expression tree.**
- **Preorder traversal is used to create a copy of the tree.**

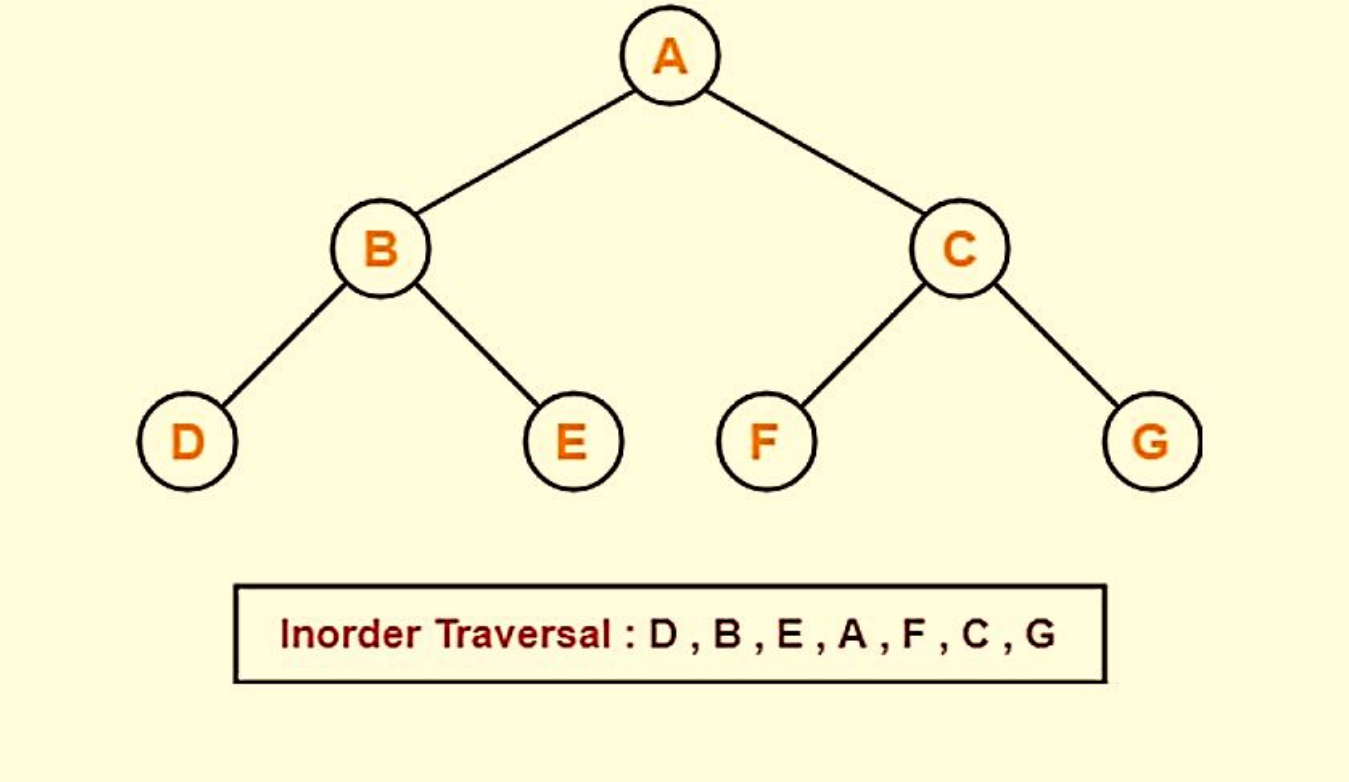
*Binary Tree Traversal (*Inorder Traversal)

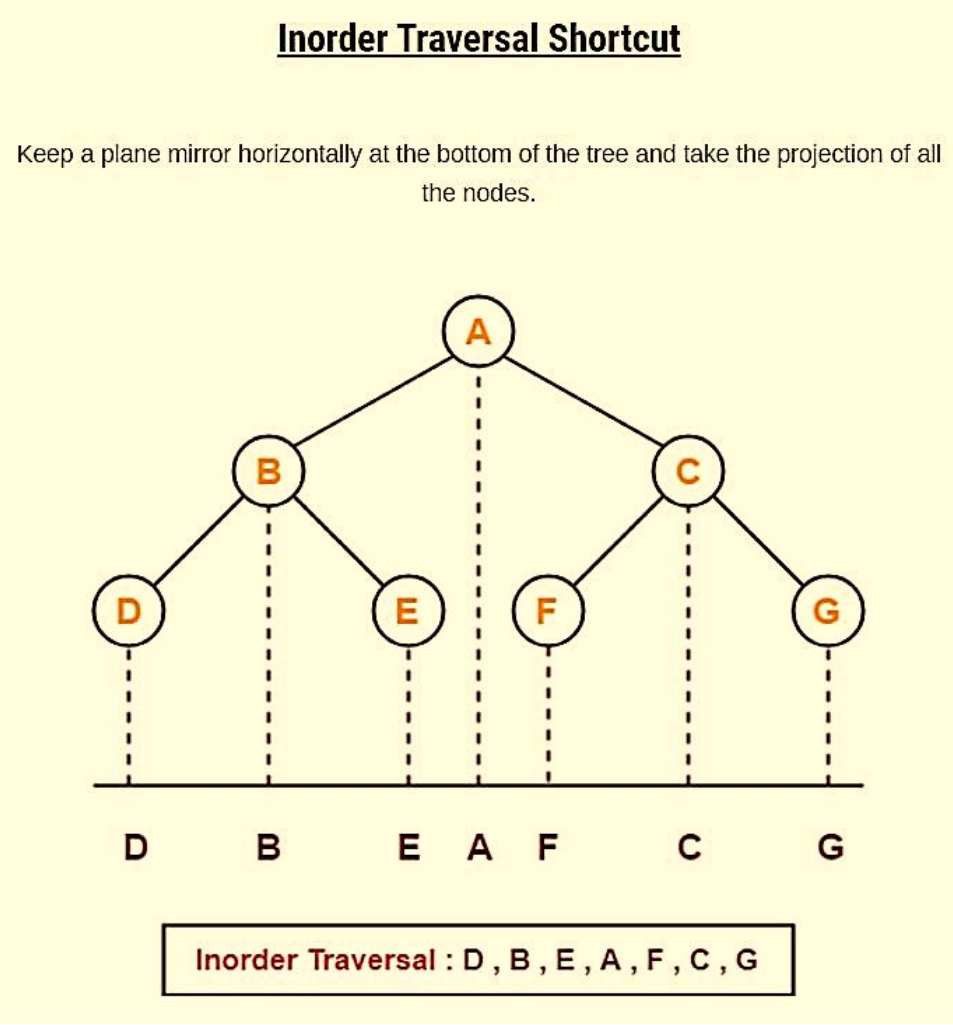
Traverse the left sub tree i.e. call Inorder (left sub tree)

Visit the root

Traverse the right sub tree i.e. call Inorder (right sub tree)

Left → Root → Right

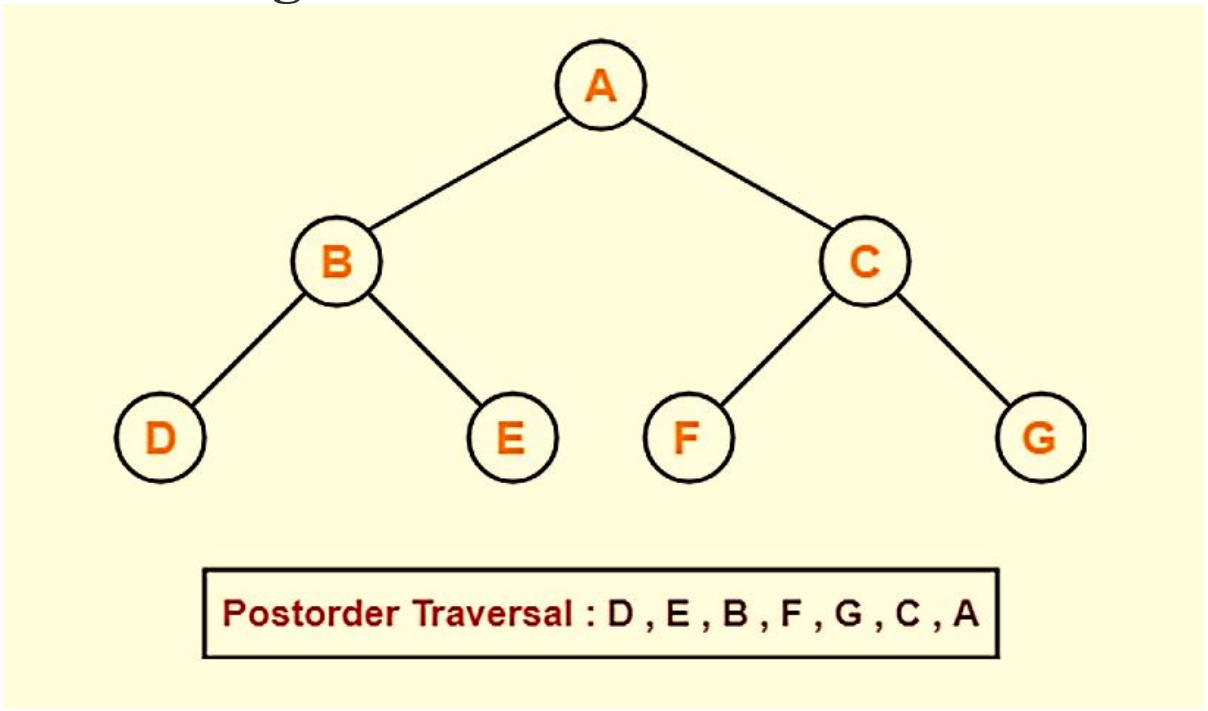




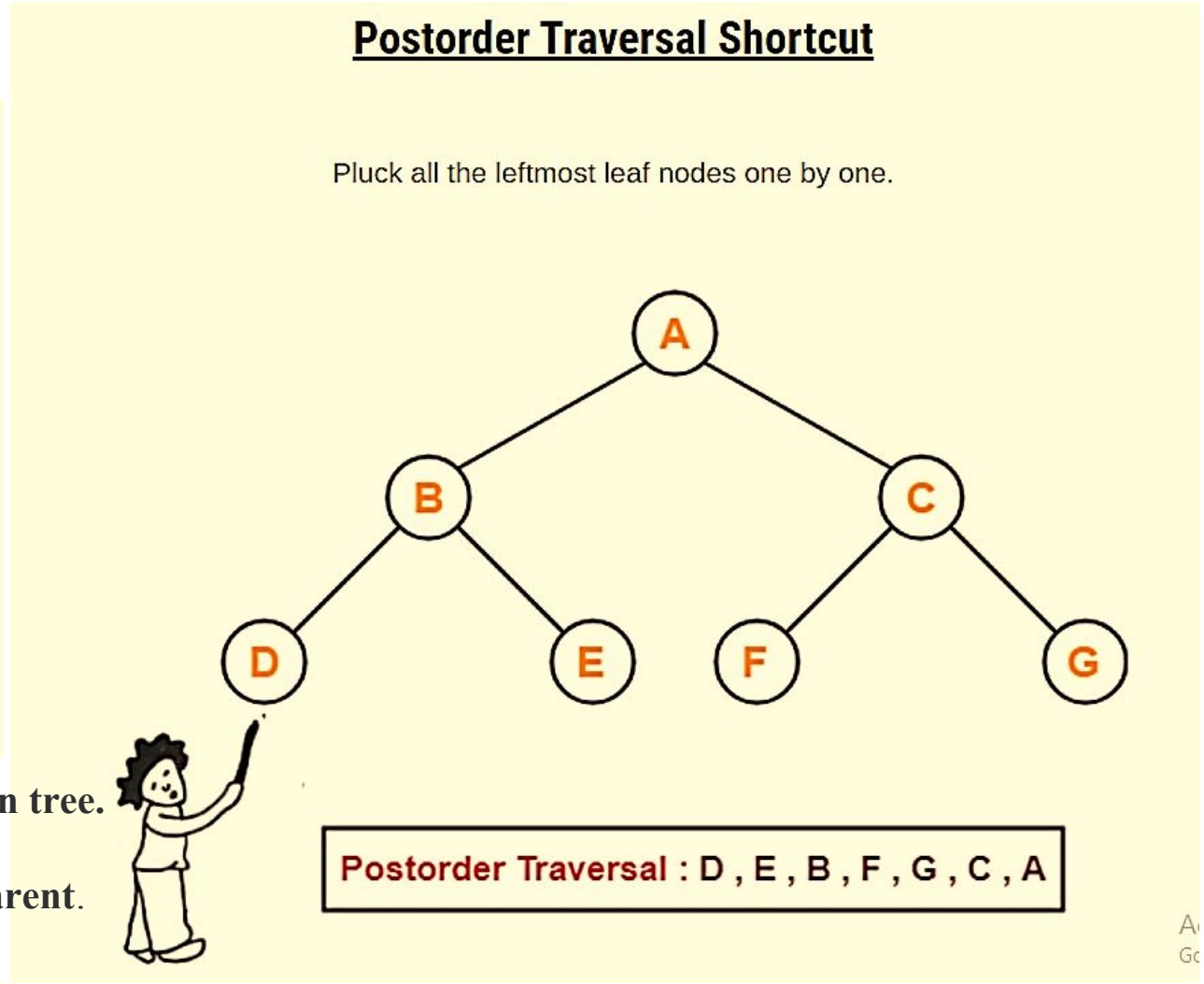
Inorder traversal is used to get infix expression of an expression tree.

*Binary Tree Traversal (*Postorder Traversal)

1.Traverse the left sub tree i.e. call Postorder (left sub tree)
2.Traverse the right sub tree i.e. call Postorder (right sub tree)
3.Visit the root
4.Left → Right → Root

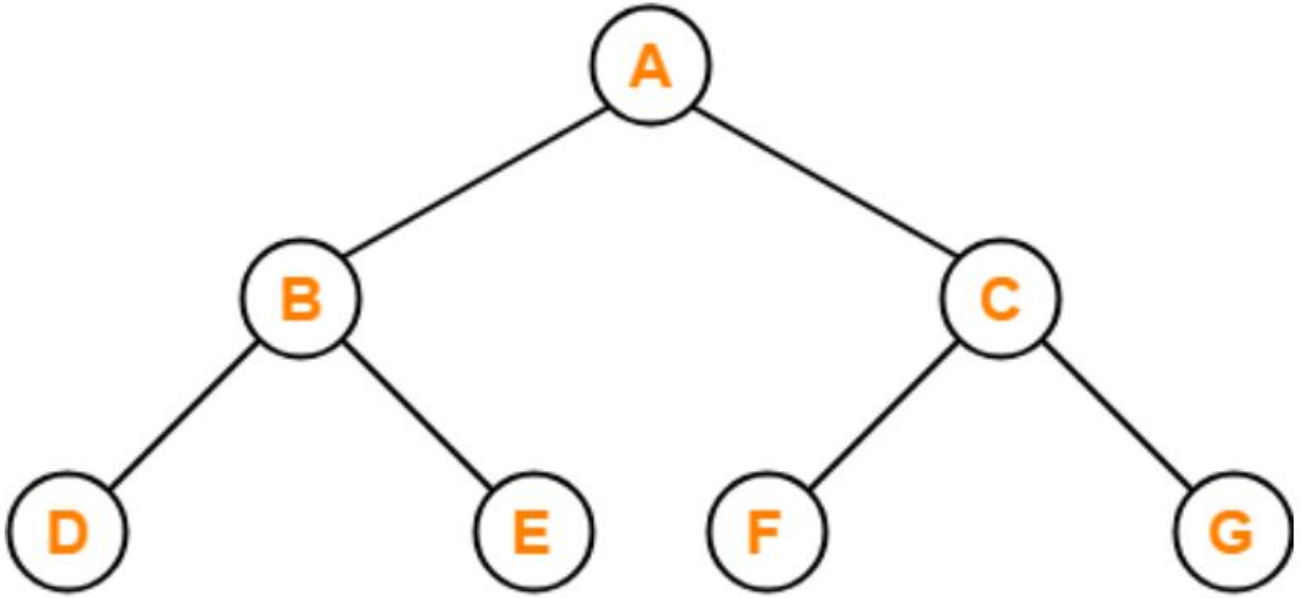


•Postorder traversal is used to get postfix expression of an expression tree.
•Postorder traversal is used to delete the tree.
•This is because it deletes the children first and then it deletes the parent.



2- Breadth First Traversal

- Breadth First Traversal of a tree prints all the nodes of a tree level by level.
- Breadth First Traversal is also called as **Level Order Traversal**.



Level Order Traversal : A , B , C , D , E , F , G

Level order traversal is used to print the data in the same order as stored in the array representation of a complete binary tree.